

Electronics Engineering Lab

Project Report

Title: WATER LEVEL INDICATOR USING BJTs (BC547)



Class: BEME-F-24-B

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Date: May 31, 2026

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Abstract:

Water level monitoring is important for preventing water wastage and ensuring the efficient use of water resources. In homes and industries, water tanks are commonly used for storage, and monitoring the water level helps avoid overflow and shortage problems. This project presents a simple Water Level Indicator using BC547 transistors, LEDs, and a buzzer. The system uses the conductivity of water to detect different water levels inside a tank. As the water level rises, corresponding LEDs glow to indicate the level of water present. When the tank becomes full, a buzzer is activated to provide an audible alert. The project is low-cost, easy to construct, and demonstrates the practical application of transistors as switching devices in electronic circuits.

Acknowledgement:

We would like to express our sincere gratitude to our instructor of Electrical Engineering Lab at Air University for providing us the opportunity to work on this project. This project helped us understand the practical application of BJTs in switching circuits. We also thank our classmates for their support and guidance throughout the development of this project.

Introduction

Water is one of the most important resources for daily life, agriculture, and industrial processes. Proper monitoring of water levels in storage tanks is necessary to avoid water wastage and ensure a continuous water supply. Traditional methods of checking water levels manually are time-consuming and may lead to overflow if not monitored regularly.

A water level indicator is an electronic system that helps monitor the amount of water present in a tank. The project developed in this report uses BC547 transistors to detect different water levels and provide visual and audio indications. LEDs are used to indicate low, medium, and high water levels, while a buzzer alerts the user when the tank becomes full.

The circuit operates on the principle of water conductivity and does not require any complex programming or expensive components. It is simple, reliable, and suitable for educational as well as practical applications.

Objectives:

The main objective of this project is to design and implement a simple and cost-effective water level monitoring system using BC547 transistors.

The specific objectives of the project are:

- To understand the working principle of BC547 transistors.
- To study the switching operation of BJTs in practical circuits.
- To detect different water levels inside a storage tank.

- To provide visual indication using LEDs.
- To provide an audible alert using a buzzer when the tank is full.
- To reduce water wastage caused by tank overflow.
- To gain practical experience in circuit design and testing.

Literature Review

Water level monitoring systems are widely used in residential, agricultural, and industrial applications to ensure efficient water management. Over the years, different techniques have been developed for measuring water levels. Traditional methods include manual inspection and float-based systems, where a floating object rises and falls with the water level. Although these methods are simple, they often require regular maintenance and may not provide accurate results.

Modern water level monitoring systems use sensors such as ultrasonic, capacitive, and conductive sensors. Ultrasonic sensors measure the distance between the sensor and the water surface using sound waves, while capacitive sensors detect changes in capacitance caused by the presence of water. These systems offer higher accuracy but are generally more expensive and complex.

The conductive probe method is one of the simplest and most economical techniques. It relies on the conductivity of water to complete an electrical circuit. When water reaches a probe, it allows a small current to flow and activate an electronic switching device. Due to its low cost, easy construction, and reliable performance, the conductive probe method is commonly used in educational and small-scale projects.

In this project, BC547 transistors are used as switching devices to detect different water levels and provide visual and audio indications. The design is simple, effective, and suitable for demonstrating practical transistor applications.

Theory of BC547 Transistor

A Bipolar Junction Transistor (BJT) is a semiconductor device commonly used for switching and amplification purposes. It consists of three terminals known as the emitter, base, and collector. The BC547 is an NPN transistor that is widely used in electronic circuits because of its low cost, availability, and reliable performance.

The operation of a BC547 transistor is based on controlling a large collector current with a small base current. When a small current is applied to the base terminal, the transistor allows current to flow from the collector to the emitter. This property makes the transistor an effective electronic switch.

The BC547 transistor can operate in three regions: cut-off, active, and saturation. In the cut-off region, the transistor remains OFF and no collector current flows. In the active region, the transistor functions as an amplifier. In the saturation region, the transistor becomes fully ON and acts like a closed switch. In this

project, the transistor is mainly operated in the saturation region to switch LEDs and the buzzer ON and OFF.

The BC547 has a maximum collector-emitter voltage of 45V and can handle collector currents up to 100mA. Due to these characteristics, it is suitable for low-power electronic applications such as water level indicators, alarm circuits, and sensor-based systems.

Working Principle:

- The Water Level Indicator works on the conductivity of water and transistor switching.
- Water contains dissolved salts and minerals that allow a small amount of electric current to flow.
- Several sensing probes are placed at different levels inside the water tank.
- One probe is connected to the positive supply, while the others are connected to the base terminals of BC547 transistors.
- When water touches a probe, it completes the electrical path and allows a small current to reach the transistor base.
- The transistor switches ON and activates the corresponding LED.
- As the water level rises, more probes come into contact with water, causing additional LEDs to glow.
- When the water reaches the highest probe, the buzzer is activated to indicate that the tank is full.
- As the water level decreases, the electrical path is broken and the LEDs and buzzer turn OFF accordingly.
- The system continuously monitors and indicates the water level inside the tank.

Principle of Operation

The working principle of the water level indicator is based on the conductivity of water and the switching action of BC547 transistors. Ordinary water contains dissolved salts and minerals that allow a small amount of electrical current to pass through it. This property is utilized to detect the presence of water at different levels.

Several probes are placed inside the water tank at different heights. One probe is connected to the positive supply, while the remaining probes are connected to the base terminals of separate BC547 transistors. As the water level rises and touches a probe, a small current flows through the water and reaches the base of the corresponding transistor.

These base current switches the transistor ON, allowing current to flow through the collector-emitter path. As a result, the connected LED lights up and indicates the current water level. When the water reaches the highest probe, the final transistor activates a buzzer, providing an audible indication that the tank is full.

The system continuously monitors the water level and automatically updates the indications as the water level changes. This simple mechanism makes the circuit reliable and easy to implement.

Components Required

- BC547 NPN Transistors $\times$ 3–4
- LEDs (Red, Yellow, Green) $\times$ 3
- Buzzer $\times$ 1
- Resistors (1k Ω , 10k Ω)
- Connecting wires
- Probes (copper wires)
- 9V Battery or DC supply
- Breadboard or PCB

Components Images

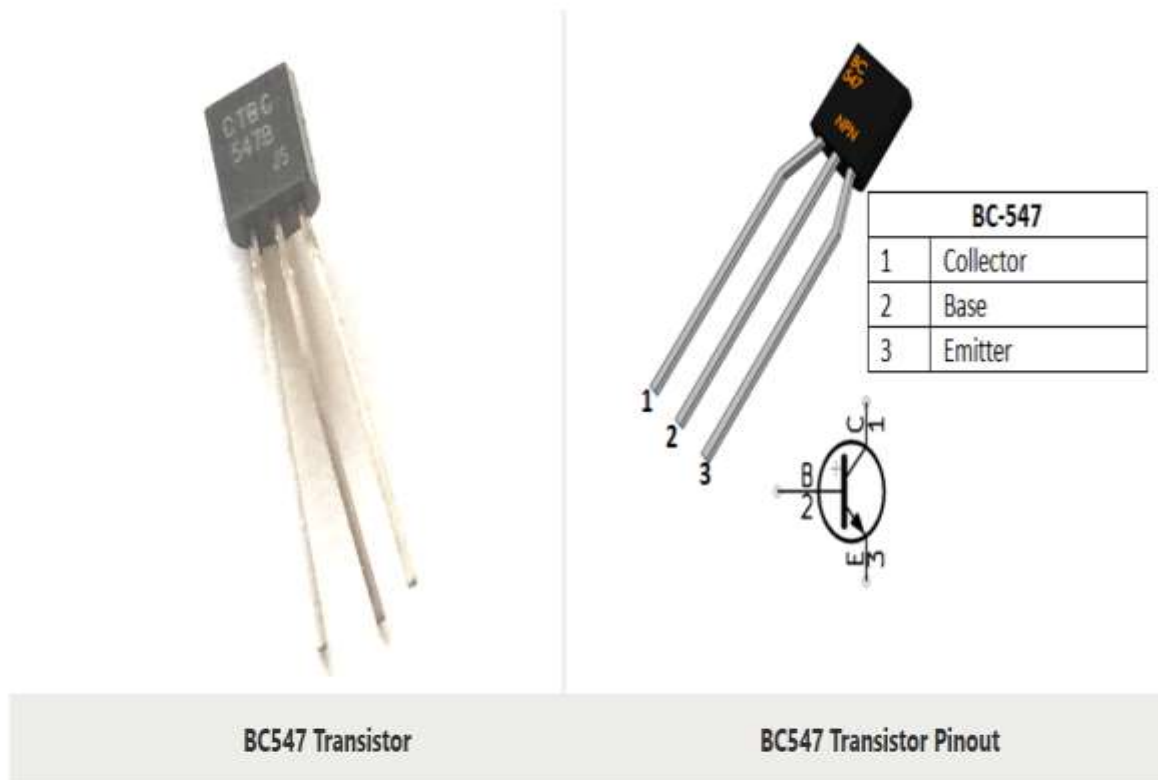


Figure 1



Figure 2



Figure 4



Figure 3

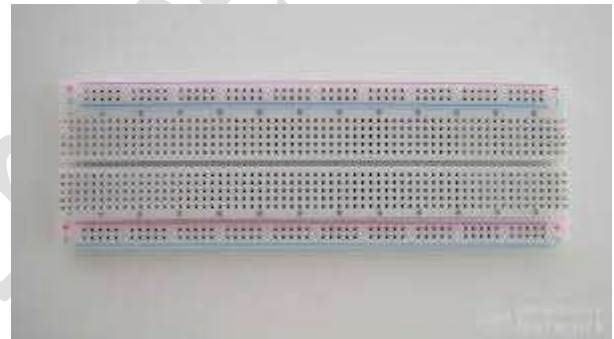


Figure 5

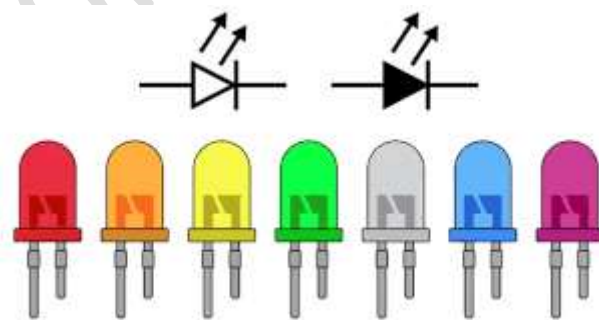


Figure 6

The diagram illustrates a 4-wire control system for a water tank. It features a 24V AC power source connected to a 4-wire terminal block. The four wires from the terminal block are connected to four 1/2 HP motors, a fan, and three indicator lights (green, yellow, and red). The wiring shows how the four wires from the terminal block connect to the motors, fan, and lights.

Figure 7

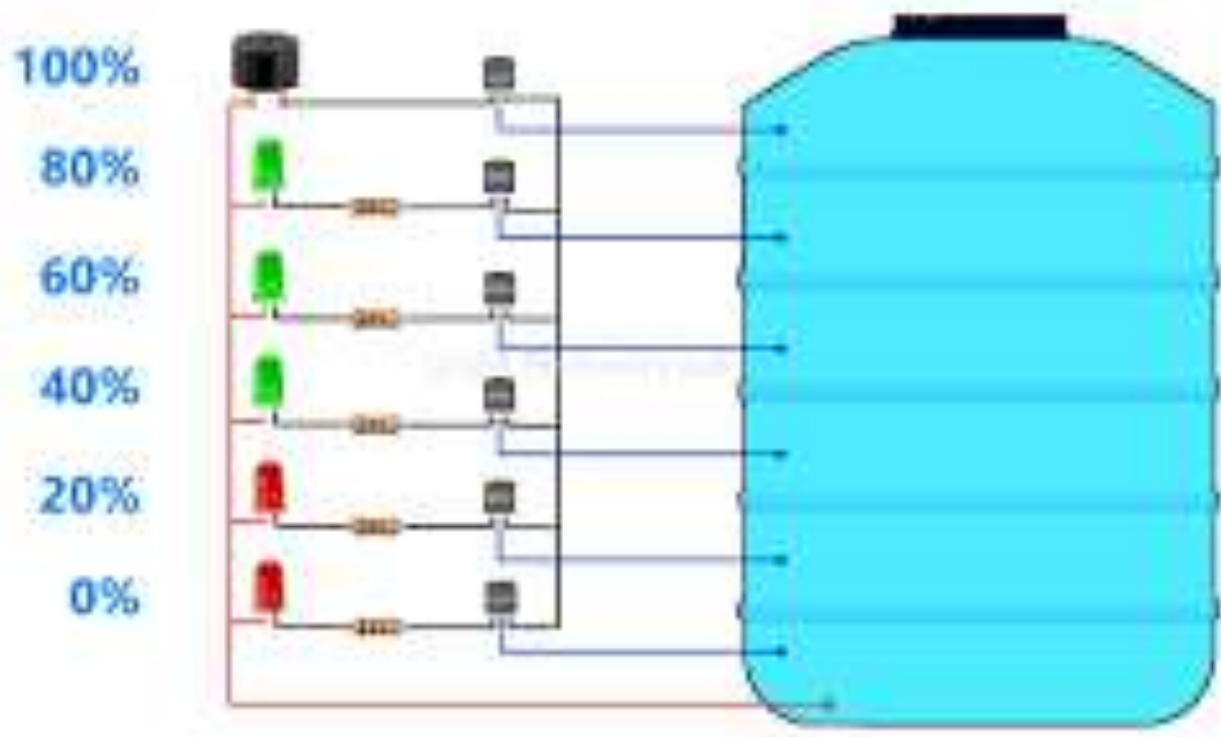


Figure 8

Description:

The figure shows a transistor-based water level indicator circuit designed to monitor different water levels inside a storage tank. Multiple sensing probes are placed at different heights inside the tank corresponding to 0%, 20%, 40%, 60%, 80%, and 100% water levels. Each probe is connected to a BC547 transistor that acts as a switch.

As the water level rises and touches a probe, a small current flows through the water and activates the corresponding transistor. The transistor then turns ON the connected LED, providing a visual indication of the current water level. Red LEDs indicate lower water levels, while green LEDs indicate higher water levels. At the maximum water level (100%), a buzzer is activated to provide an audible alert, indicating that the tank is full and preventing water overflow.

The circuit operates using the conductivity of water and provides a simple, low-cost, and effective solution for water level monitoring.

Working of Project

The project works on the principle that water can conduct electricity due to the presence of dissolved salts and minerals. Several sensing probes are placed at different heights inside the water tank. One common probe is connected to the power supply, while the remaining probes are connected to the bases of BC547 transistors.

When the tank is empty, none of the probes are in contact with water, so all transistors remain OFF and no LEDs glow. As the water level rises and reaches the first probe, a small base current flows through the transistor, causing the first LED to light up. This indicates a low water level.

When the water level continues to rise and touches the second and third probes, additional transistors are activated and the corresponding LEDs turn ON. This process continues until the water reaches the highest probe. At this stage, the final transistor switches ON the buzzer, producing an audible alarm to indicate that the tank is completely full.

As the water level decreases, the probes lose contact with water, causing the transistors to switch OFF and the LEDs to turn OFF accordingly. In this way, the circuit continuously monitors and displays the water level inside the tank.

Construction Procedure

- Collect all required components such as BC547 transistors, LEDs, resistors, buzzer, connecting wires, probes, breadboard, and a 9V power supply.
- Study the circuit diagram carefully before starting the assembly.
- Place all BC547 transistors on the breadboard according to the circuit design.
- Connect the emitter terminal of each transistor to the ground line.

- Connect suitable resistors in series with the LEDs to limit the current.
- Connect each LED to the collector terminal of its respective transistor.
- Install copper wire probes at different heights inside the water tank or container.
- Connect each sensing probe to the base terminal of the corresponding transistor.
- Connect the common probe to the positive terminal of the power supply.
- Connect the buzzer to the transistor responsible for full-level indication.
- Check all wiring connections to ensure there are no loose connections or short circuits.
- Connect the 9V battery or DC power supply to the circuit.
- Initially test the circuit with an empty tank and observe that all LEDs remain OFF.
- Gradually fill the tank with water and monitor the operation of each LED.
- Verify that the buzzer activates when the water reaches the maximum level.
- Record observations and confirm that the circuit operates according to the design requirements.

Testing and Results

Testing Procedure

The completed water level indicator circuit was tested by gradually filling a water container and observing the response of the LEDs and buzzer at different water levels. The purpose of the testing was to verify that each sensing probe correctly detected the presence of water and activated the corresponding indicator.

Observations

- When the tank was empty (0% level), all LEDs and the buzzer remained OFF.
- As the water reached the 20% probe, the first red LED turned ON, indicating a low water level.
- When the water level increased to 40%, the second LED turned ON.
- At 60% water level, the third LED illuminated, showing a medium water level.
- When the water reached 80%, the fourth LED turned ON, indicating a high water level.
- At 100% water level, the final LED and buzzer were activated, indicating that the tank was completely full.
- As the water level decreased, the LEDs turned OFF sequentially according to the water level.

Results Table

Water Level	Expected Output	Actual Output	Result
0%	All LEDs OFF	All LEDs OFF	Pass

20%	LED 1 ON	LED 1 ON	Pass
40%	LED 2 ON	LED 2 ON	Pass
60%	LED 3 ON	LED 3 ON	Pass
80%	LED 4 ON	LED 4 ON	Pass
100%	Buzzer ON	Buzzer ON	Pass

Results Discussion

The experimental results showed that the water level indicator operated successfully at all tested levels. Each transistor responded correctly when water touched its corresponding probe, and the LEDs provided clear visual indications of the water level. The buzzer was activated at the full tank condition, providing an effective warning signal. No significant delay or malfunction was observed during testing. Therefore, the circuit proved to be reliable, accurate, and suitable for water level monitoring applications.

Conclusion from Testing

The testing confirmed that the designed water level indicator successfully detected different water levels and provided the expected visual and audio outputs. The system met all project objectives and demonstrated the practical use of BC547 transistors as switching devices in water level detection circuits.

Advantages

- Low-cost system using easily available components.
- Simple circuit design and easy to understand.
- Requires minimal maintenance.
- Consumes very low power.
- Provides instant visual indication through LEDs.
- Gives an audible alert when the tank is full.
- Does not require a microcontroller or programming.
- Easy to construct on a breadboard or PCB.
- Reliable operation for household applications.
- Helps reduce water wastage due to overflow.

Limitations

- Works only with conductive water.
- Not suitable for highly purified or distilled water.
- Copper probes may corrode over time.
- Provides only level indication, not exact measurements.
- No digital display for water level monitoring.
- Requires regular inspection of probes.
- Limited range of monitoring.
- Manual control of the water pump is still required.
- Accuracy may vary depending on water quality.

Applications

- Monitoring water levels in household water tanks.
- Used in overhead and underground storage tanks.
- Water level monitoring in industrial storage systems.
- Irrigation and agricultural water management.
- Water treatment and distribution plants.
- Pump protection systems.
- Educational and laboratory projects.
- Small-scale automation systems.
- Reservoir and liquid storage monitoring.

Future Improvements

- Integration with Arduino for smart control.
- Addition of an LCD or LED display.
- Automatic ON/OFF control of water pumps.
- GSM-based SMS alerts for users.
- Wireless monitoring using Wi-Fi modules.

- IoT-based remote water level monitoring.
- Mobile application integration.
- Use of corrosion-resistant probes.
- More accurate level sensing techniques.

Solar-powered operation for energy efficiency.

Conclusion:

The Water Level Indicator using BC547 transistors is a simple and effective solution for monitoring water levels in storage tanks. The project successfully demonstrates the use of transistors as electronic switches and shows how water conductivity can be used for level detection. During testing, the system accurately indicated different water levels through LEDs and a buzzer. The circuit is economical, easy to construct, and suitable for educational as well as practical applications. Overall, the project achieved its objectives and provided valuable understanding of transistor-based control circuits.

References:

- *Basic Electronics by Floyd*
- *Electronics Lab Manuals (Air University)*
- *BC547 Datasheet (NXP Semiconductors)*
- *Online Electronics Tutorials*
- <https://www.elprocus.com/water-level-indicator-project>
- <https://youtube.com/shorts/9vCAz43ngnk?si=G6pCREJ56OjG77mT>
- <https://components101.com/transistors/bc547-transistor-pinout-datasheet>
- <https://makingcircuits.com/blog/make-this-water-level-indicator-circuit/>
- <https://www.mifratech.com/public/blogpage/Water+level+indicator+using+BC+547+Transistor>